AI-Powered Simulations for Business Negotiations: Enhancing Skill Development Through Technology

Ivana Kapráliková, Dajana Novák

Abstract: This research explores the integration of AI-powered simulations into negotiation pedagogy for non-native English-speaking business students. Conducted at the Bratislava University of Economics and Business, the study engaged 20 undergraduates in AI-assisted negotiation tasks over four weeks to examine how technology supports strategic communication and experiential learning. Using qualitative methods, reflective journals, simulation transcripts, and semi-structured interviews, the research analyzed how learners adapted language, emotional tone, and negotiation strategies in response to real-time AI feedback. Findings indicate that simulations supported strategic awareness, emotional framing, and self-regulated learning, while creating a psychologically safe environment for experimentation. Students perceived the scenarios as realistic and transferable to professional contexts, though limitations included the absence of non-verbal cues and full cultural nuance. These results highlight the potential of AI as a complement to human-led instruction, bridging theory and practice through repetitive, adaptive learning. Our research also points out institutional challenges such as limited infrastructure, large class sizes, and time constraints, which affect implementation in higher education. Overall, AI-powered simulations offer a promising pathway for curricular innovation in business communication training, provided they are embedded within hybrid models that integrate technological responsiveness with human facilitation.

Key words: AI-assisted learning, negotiation pedagogy, experiential learning, emotional intelligence, strategic communication

1 Introduction

In an increasingly globalized and digitally mediated business environment, the ability to negotiate effectively has become a core competency for professionals across industries. Negotiation now occurs across virtual platforms, diverse cultural contexts, and evolving communication norms, extending far beyond traditional face-to-face interactions. For non-native English-speaking business students, mastering negotiation discourse in English as a lingua franca presents both linguistic and strategic challenges. Conventional pedagogical approaches, lectures, case studies, and scripted role-plays, have long served as foundational tools in negotiation education (Lewicki et al., 2015). However, these methods often fail to develop the adaptability, emotional understanding, and cultural awareness needed for real negotiation situations. (Heunis et al., 2019; Zhang & Deng, 2024).

Recent scholarship increasingly highlights the potential of artificial intelligence (AI) to transform negotiation training. AI-based simulations create interactive environments where learners can engage in repeated practice, receive instant feed-

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back, and refine communication strategies in a psychologically safe setting (Dinnar et al., 2021a; Curhan, 2023). These tools are particularly beneficial for non-native speakers, offering opportunities to improve linguistic precision and strategic adaptability without the social pressure of live role-play. Moreover, AI can model complex, evolving negotiation scenarios that respond dynamically to user input, enabling learners to develop competencies such as strategic timing, emotional framing, and rhetorical flexibility (Vaccaro et al., 2025).

Despite these promising developments, the integration of AI into negotiation pedagogy remains uneven and under-theorized. While pilot programs and institutional experiments, such as those at Harvard Summer School, signal growing interest, systematic curricular adoption is still limited (Cummins & Jensen, 2024). Concerns persist regarding ethical, cultural, and emotional dimensions of AI-assisted learning. Although AI systems can analyze discourse structures and provide objective feedback, they lack the interpretive nuance and empathetic capacity of human instructors (Holstein & Aleven, 2020; Memarian & Doleck, 2023). Consequently, scholars advocate for a "*teacher – AI co-orchestration*" model, positioning AI as a complement rather than a substitute for human guidance (Kasepalu et al., 2022; Kim, 2023).

This research responds to these pedagogical and technological shifts by examining how AI-powered negotiation simulations can support the development of strategic communication skills among non-native English-speaking business students. Conducted at the Bratislava University of Economics and Business, the research engaged a culturally diverse group of undergraduates in AI-assisted negotiation tasks over a four-week period. Using qualitative methods, including reflective journals, simulation transcripts, and semi-structured interviews, the research explores how learners adapt their language, emotional tone, and strategic behavior in response to AI feedback within an experiential learning framework.

Grounded in critical discourse analysis (CDA) and contemporary constructivist pedagogy, this research emphasizes the interplay between language, power, and emotion in negotiation contexts, viewing learning as an active, socially mediated process shaped by interaction, reflection, and self-regulation. Recent scholarship positions constructivist approaches as central to higher education, highlighting collaborative learning, experiential engagement, and reflective practice as key drivers of deep learning and professional competence (Taggart & Wheeler, 2023; Woodward et al., 2020). Experiential learning, in particular, is recognized for its transformative potential, bridging theory and practice through iterative cycles of action and reflection that foster adaptability and critical thinking (Malik & Behera, 2024; Murphy & Martin, 2024). Similarly, research on self-regulated learning emphasize the importance of metacognitive strategies and autonomy in complex, technology-enhanced environments, where learners actively monitor and adjust their performance (Greene et al., 2024; Sinkkonen & Tapani, 2024). By situating AI-powered simulations within these frameworks, the research demonstrates how technology can complement, rather than replace, human-centered approaches to negotiation training.

Ultimately, it contributes to the evolving discourse on AI-assisted education by offering insights into ethical and effective integration of AI in business communication pedagogy and by addressing practical challenges faced by universities operating under resource and time constraints.

2 Literature review

2.1 Traditional and modern approaches to business negotiation education

Recent research suggests that business negotiation education is shifting from conventional models based on lectures, case studies, and static role plays to more dynamic forms of learning. Drawing on Zang and Deng (2024), traditional rational models of negotiation cannot adequately reflect emotional, cultural, and situational factors that influence real negotiations. Authors like Heunis et al. (2019) and Coleman et al. (2024) emphasise students' need for an authentic environment that allows for iteration, feedback, and safe failure – elements that traditional approaches fail to sufficiently address. These findings serve a vital purpose because they highlight the need to shift from passive acquisition of theoretical knowledge to a participatory learning model. The integration of the constructivist framework with digital technologies could bridge the gap between theory and practice and prepare students for complex interactions in intercultural contexts.

2.2 The advancement of artificial intelligence in business negotiation education

The emergence of AI and its integration within education curricula reflects the growing interest of authors in this field, highlighting the fact that AI significantly transforms the way negotiations are taught and practised (Dinnar et al., 2021; Votintseva & Johnson, 2024; Rottner, 2024; Dai et al., 2024). AI-powered simulations facilitate students' engagement with virtual agents, who reflect on their strategies, interactions, language, and emotional responses. The research of Rolf (2025) and Jiang et al. (2022) suggests that this concept of simulations with virtual agents increases engagement levels, critical thinking, and instant reflection on their own decisions. From a pedagogical perspective, drawing on Votintseva and Johnson (2024), supported by Wu et al. (2024), these simulations facilitate the development of soft skills, namely empathy, adaptability, and emotional control, which are often overlooked by traditional methods. In contrast, Bearman and Ajjawi (2023) and Lan and Zhou (2025) highlight the risk of overreliance on AI, which could lead to diminished autonomy and critical thinking when pedagogical supervision is lacking. Within the research approach adopted, it is assumed that the most effective model is represented by human-AI co-orchestration, in which AI delivers objective feedback and personalised learning, whereas the teacher provides an ethical framework, interpretation, guidance, and conceptualisation of results. This approach links analytical precision with human reflection and maintains a balance between technological innovation and human empathy in educational contexts.

2.3 Metacognition, experiental learning and AI

The current literature (Rottner, 2024; Dai et al., 2024; Trindade et al., 2025) indicates that AI-powered environments reinforce the principles of experiential learning by allowing for iterative practice, in-time feedback, and self-assessment of performance. According to Jiang et al. (2022), such environments support the cycle and depth of experience, reflection, and application of knowledge based on Kolb's experiential learning theory (1984). We consider the integration of AI within experiential learning to be key to sustainable competence development. AI tools can analyse language, strategy, and affective responses, followed by personalised learning. However, Bearman and Ajjawi (2023) point out that sole technology is not sufficient, as there is a need for teachers' reflective facilitation, which is an inevitable factor in providing ethical, intercultural, and emotional nuances of negotiation to be fully perceived and understood. This approach develops metacognitive processes crucial for long-term learning. Students learn how to think, plan a strategy, and deliberately navigate their own reactions. According to Hu and Xie (2023) and Sajida and Vijayakumar (2025), AI increases levels of self-awareness and the capacity to regulate one's own performance, which are necessary for future negotiations.

2.4 Reflective journals and feedback in AI-simulated negotiation practice

The combination of reflective journaling and AI-supported feedback system reinforces self-assessment and emotional intelligence by students (Ullmann, 2019). AI is capable of analysing text code and generating individual incentives, which support a deeper level of reflection. Simultaneously, further research (Kasepalu et al., 2022; Bearman & Ajjawi, 2023) indicates that interpretations of emotions, empathy, and intercultural context remain the privilege of the teacher. Therefore, the employment of a hybrid model is necessary, in which AI complements the analytical framework of the teacher, while the teacher facilitates the human dimension. From the empirical points of view, reflective journals prove to be an invaluable source of data on metacognitive development of students. When combined with AI analytics, it allows for not only language progress tracking, but also for emotional frameworks, self-reflection and attitudinal change.

In conclusion, it can be stated that our theoretical stance is based on the belief that effective learning occurs when AI becomes a tool for self-reflection rather than its replacement. The most effective models integrate AI, reflection, and human mediation, where technology supports analytical and critical thinking, and a learner develops values, empathy, and responsibility during the learning process.

3 Methodology

Our research employed a qualitative research design to explore the impact of AI-powered negotiation simulations on the development of negotiation skills among undergraduate business students. The research focused on participants' lived experiences, discursive behaviour, and perceptions of learning within AI-assisted training environments.

3.1 Participants

The research involved a purposive sample of 20 undergraduate students enrolled in the "Business Negotiations" course at Bratislava University of Economics and Business. All participants were non-native English speakers with advanced English proficiency, though not full mastery, representing a diverse range of linguistic and cultural backgrounds, including students from Slovakia, Ukraine, Russia, Kazakhstan, Turkey, and Colombia. This diversity provided a rich context for examining how AI-assisted negotiation training supports the development of strategic communication skills in English as a lingua franca. Participants were selected based on their willingness to engage in AI simulations and reflect on their learning experiences in English. All participants provided informed consent prior to the study. Ethical approval was obtained from the university's research ethics committee, and all data were anonymised to ensure confidentiality and participant privacy.

3.2 Data collection

Data for our research were collected over a four-week period during the spring semester of 2025. The data collection strategy in this research was grounded in a qualitative, multi-method design that allowed for both breadth and depth in understanding student development over time. Three primary sources of data were gathered: weekly reflective journals, AI-recorded negotiation simulations, and semi-structured interviews conducted post-training.

3.2.1 Reflective journals

To capture progressive findings into participants' evolving negotiation discourse and self-awareness, each student was required to maintain a **weekly reflective**

journal throughout the four-week training period. Journals were submitted electronically every Friday via Microsoft Teams, the designated platform for collecting and organising student reflections. The journal design followed a **semi-structured format**, combining open-ended prompts with targeted reflection areas. Students were encouraged to document:

- The negotiation strategies they employed during the week's simulation
- Emotional or interpersonal challenges encountered
- Their interpretation and application of AI-generated feedback
- Perceived improvements in linguistic precision, confidence, or strategic thinking

The semi-structured format ensured thematic consistency across participants while allowing for **personalised expression**, a balance supported by recent studies on reflective writing in higher education (Alt, Raichel & Naamati-Schneider, 2022; Hubbs & Brand, 2010). The journals served both as a **pedagogical tool**, supporting self-awareness in learning, and as a **qualitative data source** for cross-validating findings from simulation transcripts and interviews. Thematic analysis of journal entries focused on shifts in rhetorical awareness, emotional framing, and adaptive learning behaviours.

3.2.2 Recorded simulation sessions

Over a four-week period, each participant completed a sequence of four AI-assisted negotiation simulations using Microsoft Copilot, an adaptive digital platform selected for its capacity to provide real-time feedback, enable scenario customisation, and support session recording. Simulations were conducted weekly, with each session introducing a distinct negotiation scenario of increasing complexity and professional realism.

In week 1, students acted as purchasing officers negotiating bulk discounts from suppliers, focusing on language precision and assertive proposals. Week 2 featured internal budget negotiations between department heads, aimed at developing conditional reasoning and strategic concession-making. In week 3, students assumed the role of service providers renegotiating contract terms due to delivery setbacks, practising emotional framing and relationship management. Finally, week 4 presented a high-stakes cross-cultural joint venture negotiation, requiring rhetorical adaptability and long-term strategic thinking.

Each session was screen-recorded with participants' consent and automatically transcribed for discourse analysis. For full scenario script, see Appendix A. Tab. 1 summarizes weekly variations.

Tab. 1: *Weekly simulation scenarios and targeted negotiation competencies*

Week	Simulation scenario	Negotiation context	Targeted competencies
Week 1	*Negotiating supplier pricing*	Student represents a purchasing officer negotiating a discount for office equipment with a resistant supplier.	Assertive language use • Reducing hedging and vague phrasing • Data-supported proposals • Observing tone in influence
Week 2	*Internal budget negotiation*	Department heads advocate for budget increases in a university setting while balancing institutional constraints.	Strategic framing of offers • Conditional reasoning • Empathy and rapport-building • Strategic timing of offers and compromises
Week 3	*Service contract renegotiation*	Consultancy firm renegotiates terms with a dissatisfied client due to delays.	Emotional framing to manage tension • Rebuilding trust and rapport • Sequencing arguments for leverage • Balancing assertiveness with reassurance
Week 4	*Cross-cultural strategic venture*	Formation of a joint venture with an international partner under differing communication norms.	Cultural adaptability • Long-term strategic discourse • Rhetorical and hypothetical framing • Responding flexibly and maintaining self-control

Source: authors' own

3.2.2.1 Simulation task design and delivery

The simulation tasks were designed using a consistent yet flexible structure. Each session began with a short written briefing outlining the negotiation context, the student's role, and the primary objectives they were expected to achieve. These instructions were identical for all participants in a given week to ensure a fair starting point. However, once the simulation began, the interaction unfolded freely. No dialogue models or response templates were provided, so each student approached the negotiation in their own way, some prioritised politeness and rapport, while others focused immediately on figures and conditions. Copilot was programmed with fixed negotiation constraints such as price limits or delivery preferences, but it did not follow a scripted conversation path. It was set up by a developer at the university to keep essential rules in place, but still allow AI to respond in an open, conversational way rather than following a preset dialogue.

Assertive and well-structured proposals typically led to more cooperative replies, whereas vague or overly hesitant language triggered resistance or further questioning. This responsiveness created a sense of realism and required students to monitor their performance in real time. At the same time, the AI remained emotionally neutral, never escalating or displaying frustration. This made the simulations feel safe to experiment in, students could take risks with tone or strategy without fear of embarrassment or social tension.

Instructions were delivered through Microsoft Teams and briefly reinforced in class before each session. To avoid over-guidance, students were not given suggested phrases but were encouraged to apply rhetorical strategies previously discussed in lectures. During the simulations, researchers observed silently, stepping in only to resolve technical issues. After each session, students submitted short reflections on what they believed had worked, failed, or surprised them. This balance of minimal intervention during performance and structured reflection afterwards reflects what Holmes et al. (2021) describe as *adaptivescaffolding*, where support is carefully timed to strengthen autonomy rather than dependency.

3.2.3 Semi-structured interviews

Upon completion of the simulation training, all 20 participants took part in individual semi-structured interviews conducted by the researchers. Interviews lasted approximately 30 minutes and were held in a quiet room on campus or via secure video conferencing, depending on student preference. The interview protocol included open-ended questions exploring participants' perceptions of the AI simulations, the clarity and usefulness of real-time feedback, and the extent to which the experience influenced their confidence, adaptability, and strategic planning in negotiation contexts. All interviews were audio-recorded and transcribed for thematic analysis. Together, these three qualitative data sources provided a well-rounded understanding of how AI-powered simulations shaped students' negotiation discourse and self-perceived skill development in a higher education setting.

3.3 Data analysis

The dataset was analyzed using thematic and discourse analysis to examine how students adapted to English-language business negotiation norms. All transcripts, reflective journals, and interviews were compiled into a single dataset and coded systematically. Simulation transcripts were reviewed for discourse features outlined in the methodological framework, including hedging, modality, turn-taking, argument structure, and emotional framing. Each instance was marked and categorized by simulation stage to trace linguistic development, and cultural norms such as indirectness, deference, or assertiveness were noted to capture intercultural adjustments.

Reflective journals and interviews were analyzed inductively to identify recurring themes such as strategic awareness, confidence building, language adjustment, and intercultural adaptation. Coding involved highlighting phrases that expressed these ideas and grouping them under thematic categories. Themes were then cross-validated across journals, interviews, and transcripts to ensure consistency and credibility.

To illustrate these findings, tables were constructed to show how discourse features and strategic behaviors evolved over time, providing a clear comparison between early-stage and later-stage patterns. This approach ensured transparency and allowed both linguistic detail and learner perceptions to be integrated into the interpretation.

4 Research findings

This section presents the results of the qualitative analysis of reflective journals, AI-recorded simulation transcripts, and semi-structured interviews. The findings are organized into four interrelated categories that reflect the methodological focus: strategic awareness and language precision, emotional framing and confidence building, adaptive learning and self-regulation, and perceived realism and transferability. Each subsection integrates discourse features such as hedging, modality, turn-taking, argument structure, and emotional framing, as outlined in the methodological framework.

4.1 Strategic awareness and language precision

4.1.1 Initial patterns of language use

One of the most prominent findings was the improvement in students' ability to structure arguments and respond strategically during negotiation simulations. This development was evident across all three data sources. At the outset, participants tended to respond reactively, often using vague or overly polite language that lacked assertiveness or strategic intent.

For example:

"I think we could maybe agree on something around this price." (Participant 3, simulation transcript)

"Would you be open to discussing other options?" (Participant 11, simulation transcript)

These utterances reflect frequent hedging and modal verbs (e.g., could, might, would), which are typical of indirect or uncertain discourse.

4.1.2 Role of AI feedback in language adjustment

The AI feedback system highlighted vague language and suggested more precise alternatives, prompting students to revise their phrasing. For instance, the AI might recommend replacing "maybe agree" with:

"I propose a revised offer of €4,500, which reflects the added value we bring to the partnership."

By the third and fourth simulations, participants were observed using more structured and goal-oriented language. They began framing arguments with clear justifications, often referencing data, hypothetical outcomes, or mutual benefits:

"Given the delivery timeline and the volume we're committing to, I believe a 10% discount is a fair adjustment. This would allow us to proceed immediately and strengthen our long-term collaboration." (Participant 11)

Reflective journals confirmed this shift:

"At first, I was just reacting, but by the third simulation, I started thinking ahead – what I say now affects what I can ask for later." (Participant 7)

"The AI showed me that I was using too many uncertain phrases like 'I guess' or 'maybe.' I hadn't noticed how that made my arguments seem weaker. I started practicing clearer and more confident statements, and it really improved how I came across." (Participant 14)

4.1.3 Emergence of strategic language features

Discourse analysis revealed a reduction in hedging and filler phrases, and an increase in assertive language, conditional reasoning, and rhetorical strategies. Students began sequencing negotiation steps to manage flow:

"Let's align on the scope first. Once we have clarity there, I can offer more flexibility on pricing." (Participant 9)

They also employed value framing:

"Our offer includes post-sale support, which adds value beyond the unit price. I'd like to factor that into our discussion." (Participant 15)

And time-based leverage:

"If we finalize the agreement today, I can expedite the onboarding process by next week." (Participant 2)

Strategic pacing emerged as a key skill, with students learning to withhold concessions until leverage was established:

"I used to give away too much too soon. The AI helped me see that I needed to build my case first, then make the offer." (Participant 5)

To illustrate the discourse evolution more systematically, the Tab. 2 below contrasts early-stage and later-stage negotiation language patterns across key features identified in the methodological framework.

Tab. 2: *Evolution of strategic language use in negotiation discourse*

Discourse feature	**Early-stage pattern (Simulation 1–2)**	**Later-stage pattern (Simulation 3–4)**	**Interpretation / strategic shift**
Hedging / modality	*I think we could maybe agree on something around this price.* (P3)	*I propose a revised offer of €4,500.* (AI-suggested reformulation adopted by P11)	Reduction in uncertainty ⟶ movement toward assertiveness
Turn-taking / initiative	Reactive responses: *Okay, what do you suggest?* (P5)	Proactive structuring: *Let's align on the scope first, then discuss pricing.* (P9)	Shift from passive to controlled sequencing
Argument structure	Listing demands without justification	*Given the delivery timeline … I believe a 10% discount is fair.* (P11)	Introduction of evidence-based reasoning
Conditional reasoning	Immediate concessions: *I can lower the price if needed.*	*If we confirm delivery today, I can offer payment flexibility later.* (P13)	Emergence of trade-off logic / leverage use
Emotional framing	Formulaic politeness: *Hope this works for you.*	*I appreciate your flexibility, let's find a solution that benefits both sides.* (P7)	Emotional language becomes relational strategy

Source: authors' own

However, not all participants were able to consistently maintain strategic clarity. Several reported reverting to hedging when negotiations became complex or when AI responses were unexpected, suggesting that assertiveness was still fragile rather than fully internalised.

4.1.4 Clarity, assertiveness, and emotional framing

Several participants initially expressed uncertainty about how direct they should be when negotiating in English. Early interactions were marked by hedging and vague phrasing, which often weakened their position:

"First time I just say what I think. But then I see AI change when I speak more strong. So I try to plan better next time." (Week 2, Participant 3)

This shift toward clarity and assertiveness was reinforced by comments such as:

"When I speak too soft, AI don't give me good deal. So I try to be more clear and strong." (Participant 7)

Over time, participants reported adopting a strategy of justifying their proposals before making concessions, demonstrating growing awareness of sequencing and persuasive framing. Similarly, emotional framing evolved from formulaic politeness to purposeful empathy. At the outset, participants questioned whether their intended tone was "picked up" without non-verbal cues, noting that empathy felt scripted. With practice, they distinguished between mere politeness and strategic empathy, using reassurance to soften assertive moves and build rapport:

"I tried using more empathetic phrases before making my offer, it made the negotiation feel more cooperative." (Participant 18)

These developments reflect a growing mastery of negotiation discourse, particularly in the use of turn-taking strategies, conditional offers, and persuasive argument structure. The transition from vague, reactive phrasing to assertive, structured communication is especially significant for non-native speakers navigating English as a lingua franca.

4.2 Emotional framing and confidence building

4.2.1 Initial use of emotional language

A central finding was the development of emotional awareness and confidence among participants, particularly in how they framed their discourse during negotiation simulations. Early sessions revealed that emotional language was either absent or used inconsistently. Participants relied on formulaic politeness markers such as *"thank you"* or *"I hope this works for you"* which, while courteous, lacked strategic emotional intent. This pattern aligns with the methodological emphasis on emotional framing as a key discourse feature and pedagogical goal.

Reflective journals and interviews confirmed this initial tendency toward minimal emotional engagement. For example:

"At first, I only wrote things like 'thank you' because I thought that was enough to sound polite." (Participant 4, journal)

"I didn't use emotional phrases because I was afraid they would sound fake or too personal." (Participant 12, interview)

"In the first simulation, I focused only on numbers and forgot about tone completely." (Participant 9, journal)

4.2.2 Role of AI feedback and reflective practice

The emotionally neutral nature of the AI-powered environment provided a psychologically safe space for experimentation, allowing students to explore interpersonal dimensions without the pressure of live social interaction. AI's consistent, non-judgmental feedback encouraged reflection on tone and its impact on negotiation outcomes. This process was reinforced by journal prompts and interviews, which explicitly asked students to evaluate emotional strategies and interpersonal challenges.

4.2.3 Emergence of purposeful emotional framing

As training progressed, participants began incorporating emotionally framed statements more deliberately. These utterances served multiple functions: softening assertive proposals, building rapport, managing tension, and reinforcing collaborative intent. For example:

"I appreciate your flexibility, let's find a solution that benefits both sides." (Participant 7)

"I understand this is a critical issue for your team, and I want to make sure we address it properly." (Participant 1)

"I'm confident this proposal reflects our shared goals and long-term interests." (Participant 14)

Such statements reflect a shift from transactional to relational discourse, where emotional framing aligns interests and support cooperation, an outcome pedagogically significant given the role of emotional intelligence in business communication.

4.2.4 Reflections on emotional strategies and confidence

Reflective journals provided rich insights into evolving emotional strategies. Participant 6 wrote:

“I tried using more empathetic phrases like ’I understand your concern’ before making my offer. It made the negotiation feel more cooperative.”

Participant 17 described how emotional framing helped navigate rejection:

“When the AI rejected my offer, I didn’t panic. I used a reassuring tone and changed my proposal. That gave me confidence.”

Participant 20 emphasized emotional preparation:

“I practiced using phrases like ’I value your input’ to keep the tone respectful even when I disagreed.”

These reflections illustrate the emergence of emotional self-regulation and strategic empathy, skills essential for managing complex interpersonal dynamics.

4.2.5 Discourse features and behavioral adjustments

Later transcripts confirmed these trends. Students used emotional language to influence tone and guide negotiation flow. Participant 3 noted:

“When the AI challenged me, I changed how I said things and tried to be more empathetic. That shift in tone helped me steer the conversation back in a better direction.”

Participant 8 described a shift from avoidance to engagement:

“I used to avoid conflict, but now I see that being calm and direct helps more than being overly polite.”

Interviews revealed growing awareness of non-verbal cues and tone control, even in text-based simulations. Several participants practiced speech modulation, strategic pausing, and tone adjustments to convey calmness and authority. For example:

“I started paying attention to how my voice sounded. I slowed down and added pauses to sound more confident.” (Participant 13, interview)

“Before, I wrote messages quickly without thinking about tone. Now I check if my words sound reassuring, not just factual.” (Participant 16, interview)

To provide a structured overview of how emotional framing evolved throughout the simulations, Tab. 3 below contrasts early and later-stage discourse behaviours across key affective features.

Tab. 3: *Development of emotional framing and confidence*

Emotional / interpersonal feature	Early-stage pattern (Simulation 1–2)	Later-stage pattern (Simulation 3–4)	Interpretation / strategic shift
Emotional expression	Minimal politeness: *Thank you … I hope this works.* (P4)	Purposeful reassurance: *I understand this is important for your team. Let's find a fair solution.* (P1)	From surface politeness to strategic empathy
Tone awareness	Neutral phrasing: *Here is my offer.* (P9)	Softened assertiveness: *I'm confident this proposal reflects our shared goals.* (P14)	Tone used as persuasive tool
Response to rejection / pressure	Quick concession: *Okay, we can change that.* (P12)	Emotional recovery: *I see your concern, what if we adjust this instead?* (P17)	Emergence of resilience and reframing
Confidence in delivery	Hesitant phrasing: *Maybe we can try something else?* (P12)	Decisive openings: *I value your input. To move forward, I suggest …* (P18)	Growth in self-assurance
Intercultural balance	*I was afraid emotion would sound fake.* (P12)	*I learned to use emotion to connect and convince.* (P19)	Emotional strategy refined across cultures

Source: authors' own

4.2.6 Intercultural dimensions of emotional framing

Cultural background influenced emotional strategies. Participants from expressive cultures (e.g., Colombia) and reserved cultures (e.g. Eastern Europe) reflected on balancing cultural instincts with English-language business norms.

"I usually speak with a lot of emotion, but I learned that using emotion in a smart way can help not only to connect with people, but also to convince them." (Participant 19, Colombia)

"In Slovak, we're taught to be quite formal and reserved. At first, I didn't think showing emotion was necessary, but the AI made me realize that using phrases like 'I understand your concern' actually helped me sound more professional." (Participant 6, Slovakia)

"I was used to keeping things very neutral and direct, but I saw that in English negotiations, it's important to show some emotion to build trust." (Participant 10, Kazakhstan)

These reflections highlight the intercultural negotiation of emotional norms and the role of AI simulations as culturally neutral spaces for experimentation.

Overall, emotional framing emerged as a critical component of negotiation discourse, shaped by AI feedback, reflective practice, and intercultural awareness. Participants learned to use emotional language purposefully, regulate tone, and adapt strategies to maintain control and build trust. These developments contribute to a holistic understanding of negotiation competence, integrating linguistic precision with emotional and interpersonal agility.

4.3 Adaptive learning and selfregulation

4.3.1 Overview and learning environment

The research revealed that the AI-powered simulations supported the development of adaptive learning behaviours and self-regulation among participants. These developments were particularly evident in how participants monitored their performance, responded to feedback, and adjusted their negotiation strategies over time. The dynamic responsiveness of the AI system, its ability to adapt to the structure, tone, and timing of student input, created a learning environment that encouraged reflection-in-action and improvement.

4.3.2 From passive participation to active learning

From the early stages of the training, participants began to recognize that the AI was not simply reacting to their words, but to how they framed their arguments, sequenced their offers, and managed conversational flow. This realization prompted a shift from passive participation to active learning. Participant 4 described this transition:

"First time I just say what I think. But then I see AI change when I speak more strong. So I try to plan better next time."

Participant 10 similarly noted:

"When I speak too soft, AI don't give me good deal. So I try to be more clear and strong. It works better."

This trial-and-error learning was common across the dataset. Participants used AI feedback and scenario shifts as cues to evaluate the effectiveness of their language and tactics. Over time, they developed a more nuanced understanding of cause and effect in negotiation discourse, how specific phrasing, tone, or timing influenced the AI's responses and the overall direction of the negotiation. Simulation transcript illustrate this progression:

"If we agree on the delivery schedule today, I can offer flexibility on pricing later." (Participant 13, Simulation 3)

The following two examples were extracted from interviews, showing how participants reflected on their evolving strategies:

"I realized that starting with a clear proposal makes AI respond faster, so now I say: 'I propose €4,800 with delivery in 10 days." (Participant 6, interview)

"Before, I just waited for AI to suggest something. Now I start with: 'Let's align on scope first, then discuss price.' It changes the whole flow." (Participant 9, interview)

These examples confirm that learners moved from instinctive, reactive responses toward deliberate, structured openings and sequencing strategies, demonstrating active engagement with the negotiation process.

4.3.3 Discourse-level shifts and strategic planning

Discourse analysis confirmed a shift from reactive responses to planned, strategic discourse. By the fourth simulations, many participants began to delay concessions, use conditional offers, and revisit earlier points to strengthen their position. For example:

"If we can agree on the delivery schedule today, I'm open to discussing the pricing flexibility next week." (Participant 13)

Additional examples from later simulations illustrate this trend:

"Let's finalize the scope first. Once that's clear, I can offer more flexibility on payment terms." (Participant 11, simulation transcript)

"If you can confirm the warranty extension, I'm prepared to adjust the unit price accordingly." (Participant 7, simulation transcript)

"Before we move to pricing, I'd like to ensure the delivery timeline meets your requirements – then we can discuss discounts." (Participant 15, simulation transcript)

These excerpts demonstrate how learners began to structure negotiations around sequencing and conditional reasoning, signaling a deliberate shift toward strategic planning rather than reactive engagement.

4.3.4 Metacognitive strategies and performance tracking

Students also developed metacognitive strategies to track and refine their performance. Participant 14 wrote:

"After each session, I looked at where I lost control of the conversation. That helped me change my approach next time."

Participant 7 reflected on adapting to the AI's tone:

"When the AI was more direct, I realized I needed to organize my ideas better. I started using clear openings and summaries to make my message stronger."

Participant 20 described using feedback loops to improve negotiation flow:

"I noticed that when I mirrored the AI's phrasing, the negotiation flowed better. So I started doing that more."

These reflections illustrate the emergence of self-regulated learning, where participants not only responded to feedback but internalized it as part of their strategic development. The AI's consistent and emotionally neutral responses created a low-pressure environment that supported experimentation and encouraged participants to take ownership of their progress. To consolidate these observations, Tab. 4 provides an overview of how participants' adaptive behaviours shifted from reactive responses to strategic self-regulation across the simulation process.

Tab. 4: *Development of adaptive learning and self-regulation*

Adaptive behaviour feature	Early-stage pattern (Simulation 1–2)	Later-stage pattern (Simulation 3–4)	Interpretation / strategic shift
Response strategy	Reactive replies: *I wait for AI to suggest something.* (P9)	Proactive openings: *Let's clarify the scope first.* (P11)	From passive following to active steering
Adjustment to AI feedback	Ignoring resistance or conceding quickly	*If delivery is an issue, I can adjust pricing after we align.* (P13)	Reflective trial-and-error behaviour
Planning and anticipation	*I just say what I think.* (P4)	*I practiced in my head how AI might react.* (P16)	Growth in metacognitive awareness
Emotional self-regulation	*I didn't know what to do.* (P12)	*I stopped, thought, and tried a different approach.* (P8)	Controlled emotional response
Cultural adaptation	Relying on native norms	*I slowed down to match business tone.* (P18)	Intercultural flexibility

Source: authors' own

4.3.5 Emotional selfregulation and strategic foresight

Participants also became more aware of their emotional responses and learned to regulate them during challenging moments. Participant 8 shared:

"Sometimes I feel nervous when AI says no. But I stop, think again, and try different way. It helps me not to give up."

Participant 16 described mental rehearsal as a preparation strategy:

"Before last simulation, I practice in my head what I want to say. I think about what AI maybe answers. It helps me feel ready."

These behaviours reflect a growing capacity for emotional self-regulation and strategic foresight, skills that are essential for professional readiness in high-stakes communication contexts.

4.3.6 Intercultural influences on adaptive learning

Cultural background influenced how students approached adaptive learning. Learners from different cultural contexts brought distinct expectations about negotiation structure, authority, and flexibility. A Slovak student (Participant 12) reflected:

"In our culture, we're used to following rules and being quite formal. At first, I thought I had to stick to one strategy. But the AI showed me that adapting and changing tactics is actually more effective. I started experimenting more and it helped me feel more confident."

A Kazakh participant (Participant 18) described how they adjusted their pacing and tone based on cultural norms and AI feedback:

"I usually try to be very direct and fast in negotiations, but I saw that in English it's better to slow down and explain more. The AI helped me see that timing and tone really matter. I started planning my responses and waiting before making offers."

These reflections highlight how adaptive learning is not only cognitive but also culturally situated. Students had to reconcile their native communication styles with the expectations of English-language business discourse. The AI simulations provided a culturally neutral space for this negotiation, allowing learners to test strategies without fear of social missteps or hierarchical constraints.

4.3.7 Additional reflections on strategy development

Additional reflections further illustrate the diversity of adaptive strategies. A Ukrainian student (Participant 9) noted:

"I used to think that being polite and waiting was the best way. But I learned that sometimes you need to take initiative and be more assertive. I practiced making stronger openings and it changed how the AI responded."

A Turkish participant (Participant 15) shared:

"I realized that negotiation is not just about what you say, but when and how you say it. I started using pauses and summaries to guide the conversation. It helped me feel more in control."

These examples demonstrate how students engaged in reflective practice and strategic experimentation, supported by the AI's adaptive feedback mechanisms. The simulations encouraged learners to test hypotheses about negotiation behaviour, observe outcomes, and refine their approach, a process that aligns with constructivist pedagogy and supports the development of autonomous learning.

In summary, the AI-powered simulations provided a rich environment for adaptive learning and self-regulation. Participants moved from instinctive responses to deliberate, strategic communication, supported by real-time feedback and reflective tools. They developed personalized strategies for planning, pacing, and emotional control, and many began to track their own progress across sessions. These developments are particularly valuable for non-native English speakers navigating intercultural business discourse, as they foster both linguistic competence and strategic agility. The findings prove the pedagogical potential of AI-assisted environments in cultivating reflective, self-directed learners equipped for complex negotiation tasks.

4.4 Perceived realism and transferability to professional contexts

4.4.1 Perceptions of realism and professional relevance

The research demonstrated that participants consistently perceived the AI-powered negotiation simulations as realistic, professionally relevant, and transferable to real-world contexts. Despite knowing they were interacting with an artificial agent, participants described the scenarios as authentic rehearsals for workplace communication. This perception of realism played a critical role in enhancing learner engagement, motivation, and confidence, aligning with the experiential learning framework that underpins the research.

Participant 18 captured this sentiment shortly:

"It felt like a rehearsal for real life. I now feel more ready to negotiate with clients or partners."

4.4.2 Evidence of skill transfer to real contexts

Reflective journals and interviews revealed multiple instances where students applied skills developed during the simulations to professional or academic settings. Participant 7 wrote:

"In my internship, I had to talk to a supplier. I remembered how I used framing in the AI task and tried the same. It helped me stay calm and clear."

Participant 2 noted:

"Before my interview, I practiced how to present my ideas clearly, just like in the AI sessions. It made me feel more confident."

Participant 15 described applying simulation strategies in a group project:

"I used the same opening strategy from the simulation to start our team pitch. It helped set a professional tone."

Additional examples from simulation transcripts illustrate the professional tone students adopted:

"Before we move to pricing, I'd like to understand your delivery expectations. That will help us align our offer more closely with your needs." (Participant 12, Simulation 4)

"If we can confirm the warranty terms today, I'm ready to finalize the pricing adjustments immediately." (Participant 9, Simulation 4)

"Let's agree on the timeline first, then we can discuss the cost implications." (Participant 6, Simulation 3)

These reflections and examples illustrate the transfer of learning from the simulation environment to real-life communication tasks. Participants internalized strategic discourse patterns and emotional framing techniques, applying them in contexts requiring clarity, persuasion, and professionalism.

4.4.3 Role of unpredictability and strategic adaptation

Participants also valued the unpredictability of the AI's responses, which mimicked the variability and ambiguity of real negotiation dynamics. Participant 5 explained:

"With AI, I never know exactly what it will say. It's like a real person. I have to listen carefully and think fast."

This unpredictability was not perceived as a limitation but rather as a strength of the learning environment. It required students to adapt in real time, manage uncertainty, and respond strategically, skills essential in professional negotiations. Additional examples illustrate this adaptive process:

"Sometimes the AI changed the topic suddenly, so I had to reorganize my argument on the spot." (Participant 8, interview)

"When AI rejected my offer without explanation, I learned to ask clarifying questions instead of giving up." (Participant 14, interview)

"In one simulation, AI proposed an unexpected delivery term. I had to quickly adjust my pricing strategy and explain why it was still fair." (Participant 11, simulation transcript)

These excerpts confirm that unpredictability fostered flexibility and strategic thinking, reinforcing the pedagogical value of dynamic, AI-mediated negotiation environments. The following Tab. 5 consolidates key indicators of perceived realism and transferability reported across journals, interviews, and simulation transcripts.

Tab. 5: *Perceived realism and transferability to professional contexts*

Realism and transfer feature	Observed perception / behaviour	Interpretation
Sense of realism	*It felt like rehearsal for real life.* (P18)	AI perceived as credible proxy
Application to real tasks	*I used framing from AI simulations in my internship call.* (P7)	Skill transfer confirmed
Handling unpredictability	*AI changed topic suddenly – I had to reorganise quickly.* (P8)	Authentic negotiation pressure
Intercultural awareness	*I had to speak more directly than I'm used to.* (P15)	Negotiating cultural identity vs. global norms
Recognition of AI limits	*Lacked emotion, but that made it easier to experiment.* (P11)	Neutrality as both limitation and benefit

Source: authors' own

4.4.4 Intercultural dimensions of realism and transferability

Cultural background influenced how students interpreted and responded to the realism and transferability of the simulations. Participants from different cultural contexts brought distinct expectations about what constitutes "realistic" communication and how transferable skills should be framed.

A Slovak student (Participant 14) reflected:

"I was surprised how real the AI felt. It didn't just follow a script, it reacted to what I said. That made me think more carefully about how I structure my arguments, like I would in a real meeting. It helped me prepare for situations where I need to speak clearly and confidently, even if I'm nervous."

A Kazakh participant (Participant 17) shared:

"In Kazakhstan, negotiations are usually quite formal and follow a strict hierarchy. The AI was more relaxed, so I had to change how I communicated. It taught me to be more flexible and quick to respond. I think this will really help when I work with people from other countries."

A Turkish student (Participant 15) added:

"In my country, we often communicate in a more indirect way and show a lot of respect for authority. But in the AI simulations, I had to speak more directly and take the lead. It felt a bit uncomfortable at first, but now I understand how useful this skill is in international business."

These reflections highlight the intercultural dimension of perceived realism and transferability. Students were not only practicing language skills but also negotiating cultural expectations and adapting to global business norms. The AI simulations provided a culturally neutral space where learners could experiment with different styles and strategies without fear of misinterpretation or social risk. This process supported the development of intercultural communicative competence, one of the objectives of our research.

4.4.5 Perceived limitations and pedagogical value

Some participants noted limitations in the emotional and contextual complexity of the simulations. While the AI provided consistent and structured feedback, it lacked non-verbal cues, spontaneous emotional reactions, and deeper cultural nuance. However, this emotional detachment also created a low-pressure environment that encouraged experimentation and self-observation, particularly valuable in the early stages of skill development.

Participant 11 commented:

"It was easier to try new things with the AI because I didn't feel judged. I could test different tones and strategies and see what worked."

Participant 8 added:

"I know it's not a real person, but the way it responded made me think about how I would handle a real negotiation. It helped me prepare mentally."

These insights suggest that the perceived realism of the simulations was not dependent on perfect replication of human interaction, but rather on the relevance, responsiveness, and strategic challenge they presented. The simulations created a bridge between classroom learning and professional application, allowing students to rehearse, refine, and transfer their skills in a psychologically safe yet professionally authentic environment.

In sum, the AI-powered simulations were not only effective in developing communicative competence but were also perceived by students as realistic, transferable, and professionally valuable. This perception significantly contributed to learner motivation, engagement, and confidence, the key affective factors in successful language acquisition and workplace readiness. The findings affirm the pedagogical potential of AI-assisted environments in preparing non-native English-speaking business students for real-world negotiation tasks across diverse cultural and professional contexts.

While most participants demonstrated strategic and emotional growth, some expressed skepticism about AI as a negotiation partner. A few reported frustration when the system failed to recognize humor, indirect hints, or culturally embedded politeness strategies, while others noted moments of over-adaptation, prioritizing responses they believed the AI would reward rather than maintaining authentic communication styles. These limitations stem from the constructed nature of AI-driven interactions, which rely on programmed constraints and linguistic algorithms and lack genuine human emotion or cultural nuance.

Although the simulations successfully mimicked negotiation dynamics and provided responsive feedback, they could not replicate the full complexity of human interaction, such as spontaneous emotional reactions, non-verbal cues, and culturally embedded rituals. This raises questions about the depth of experiential learning in AI-mediated environments. The sense of realism reported by students may reflect the structured challenge and psychological safety of the simulations rather than their fidelity to real-world interpersonal negotiation.

Pedagogical implications

The findings highlight the pedagogical potential of AI-powered simulations as a complementary tool rather than a replacement for traditional instruction. Their adaptive feedback creates a psychologically safe environment for experimentation, enabling learners to rehearse strategic discourse without fear of social judgment. This aligns with constructivist and experiential learning principles, where iterative practice and reflection foster deeper skill acquisition.

AI simulations effectively bridge the gap between theory and practice by allowing students to operationalize persuasive strategies in dynamic, interactive scenarios. Through repeated cycles of practice, feedback, and reflection, learners internalized negotiation techniques and developed emotional awareness alongside linguistic precision. Experimentation with tone, rapport-building, and reassurance strategies supported the growth of emotional intelligence, a competency increasingly valued in global business communication.

Another key implication is the promotion of learner autonomy and self-regulation. Adaptive feedback encouraged metacognitive skills, prompting students to monitor performance, evaluate outcomes, and adjust strategies proactively. Combined with reflective journaling and interviews, the simulations created a robust framework for developing independent, adaptive communicators prepared for high-stakes professional interactions. The culturally neutral AI environment also provided a safe space for reconciling native communication norms with global business expectations, enhancing intercultural competence.

To maximize these benefits, future curricula should adopt hybrid models that integrate AI simulations with instructor-led debriefings. Such integration ensures that emotional nuance, ethical reasoning, and cultural interpretation, elements beyond AI's scope, remain central to negotiation training. Reflective scaffolding through journals and peer feedback should accompany simulations, and scenario design should progress in complexity to mirror real-world challenges. While AI offers realism and adaptability, its limitations in replicating non-verbal cues and spontaneous emotional dynamics call for multimodal enhancements and hybrid approaches that merge technological scalability with human empathy. Longitudinal research should further examine the transferability of these skills to professional contexts.

AI-powered simulations represent a promising pathway for innovation in negotiation pedagogy. By combining technological responsiveness with reflective practice and human guidance, educators can create learning environments that are adaptive, culturally sensitive, and ethically grounded.

Conclusion and recommendations

This research examined the role of AI-powered simulations in enhancing negotiation pedagogy for non-native English-speaking business students, situating the discussion within the broader context of technology-driven learning. While the research confirms the pedagogical potential of AI-assisted environments, their implementation in higher education remains constrained by structural and institutional realities. At Bratislava University of Economics and Business, for example, the integration of such tools is challenged by limited technological infrastructure, large class sizes, and time restrictions. Weekly seminars of ninety minutes leave little room for iterative practice and reflective debriefing, both of which are essential for developing strategic and emotional competencies in negotiation. These constraints are not unique to our institution but reflect systemic issues across many universities where resource allocation and curricular flexibility lag behind technological innovation.

Beyond logistical limitations, the research raises important questions for future discussion. How can universities ensure that AI complements rather than replaces human judgment, particularly in areas requiring cultural sensitivity and ethical reasoning? What frameworks should govern the responsible use of AI in educational settings, addressing concerns of bias, transparency, and data privacy? These questions call for interdisciplinary collaboration among educators, technologists, and policy makers to establish standards that balance innovation with pedagogical integrity.

To address these challenges, several practical recommendations emerge. First, institutions should consider **incremental adoption strategies**, such as integrating short AI-based negotiation tasks alongside traditional role-plays, rather than attempting full-scale implementation immediately. This blended approach allows educators to maintain human oversight while leveraging the adaptive feedback and scalability of AI systems. Second, **investment in infrastructure** is critical. Universities need reliable hardware, software licenses, and technical support to ensure that AI tools function effectively in classroom settings. Third, **curricular redesign** should prioritize flexibility, incorporating asynchronous learning components that extend practice beyond the constraints of weekly seminars. Online platforms can host AI simulations and reflective activities, enabling students to engage with negotiation tasks at their own pace while preserving valuable in-class time for discussion and analysis. Finally, **faculty development programs** are essential to equip instructors with the skills to interpret AI feedback, contextualize it within cultural and ethical frameworks, and guide students toward reflective, strategic learning.

In conclusion, AI-assisted negotiation training offers a promising pathway for innovation in business communication education, but its success depends on systemic adaptation rather than technological enthusiasm alone. Future research should explore hybrid models that combine AI responsiveness with human facilitation, investigate multimodal enhancements to capture non-verbal and emotional dynamics, and assess the long-term transferability of skills to professional contexts. By addressing infrastructural, pedagogical, and ethical considerations, universities can move toward a model of negotiation education that is both technologically advanced and deeply human-centered.

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Appendix A

To illustrate how this pedagogical model operates in practice, the following scenario was implemented using Microsoft Copilot as the AI negotiation partner:

Scenario title: Negotiating a supply contract with a new client

Student role: Sales representative at TechNova Components

AI Role: Procurement Manager at SmartEdge Devices

Scenario brief:

You are negotiating a 12-month supply contract. Your goals are to:

- Secure a minimum order of 10,000 units per quarter
- Maintain a price of €4.50 per unit
- Offer standard delivery terms (3 weeks)

The client may:

- Request a lower price
- Ask for faster delivery
- Seek flexible order quantities

Instructions to students:

Use persuasive language, emotional framing, and strategic concessions to reach a mutually beneficial agreement. Reflect on your performance after the simulation.

Reflection prompts:

- What strategy did you use to persuade the client?
- How did you respond to pressure or objections?
- Did you use emotional language? How did it affect the negotiation?
- What would you do differently next time?

Example interaction:

Student: *"Thank you for meeting with me. We're confident that our components will meet your performance expectations. For a 12-month contract, we propose a price of €4.50 per unit, with a minimum order of 10,000 units per quarter."*

AI: *"Thanks for the proposal. That price is a bit above our current budget. We were hoping for something closer to €4.20 per unit. Is there any flexibility?"*

Student: *"I understand your concern. Our pricing reflects the reliability and long-term durability of our components, which could reduce your maintenance costs. If you're open to increasing the quarterly volume to 12,000 units, we could explore a small discount."*

AI: *"That's a possibility. What about delivery times? We're currently facing tight deadlines and would need faster turnaround than three weeks."*

Student: *"Standard delivery is three weeks, but we can offer expedited shipping for priority orders at a premium. Would that help meet your deadlines without compromising the overall cost structure?"*

AI: *"It might. Let me ask, how flexible are you on order quantities? We sometimes experience fluctuations in demand."*

Student: *"We can offer some flexibility, but we'd need a guaranteed minimum to ensure production efficiency. Perhaps we could agree on a base quantity with the option to adjust within a 10% range?"*

AI: *"That sounds reasonable. I'll need to review this internally, but your proposal is well-structured."*

Authors

Mgr. Ivana Kapráliková, PhD. holds an MA in English Language and Education, an MA in Political Science, and a PhD. She is an Assistant Professor of Business English at the Bratislava University of Economics and Business and at the Faculty of Economics of Pan-European University in Bratislava, Slovakia, where she teaches Business English and Business Negotiations. Her current research is situated at the intersection of critical discourse analysis and applied linguistics, with a primary focus on the political rhetoric of Central European politicians. In parallel, she explores the role of language in business negotiations, emphasizing the development of graduates' soft skills and the need to bridge theory and practice in higher education. Her recent work examines business negotiation discourse through corpus-based analyses and investigates the use of AI-driven simulation models as innovative tools in teaching business negotiations. She is the project leader of the KEGA project focusing on innovative methods in teaching Business Negotiations in English to enhance graduate employability.

Mgr. Dajana Novák is a lecturer at the Bratislava University of Economics and Business, where she teaches Business English and Business Negotiations. She is also a PhD student at the Faculty of Applied Languages, where she previously earned her Master's degree in Foreign Languages and Intercultural Communication. Her current doctoral research focuses on Persuasive Strategies in AI-Generated Negotiation Discourse, with emphasis on hedging, question strategies, and cultural frames of persuasion. She investigates how artificial intelligence simulates different cultural communication styles in negotiation settings and compares these patterns with the discourse of human negotiators. Her research explores pragmatic indicators of uncertainty, functional types of questions, and the sequential organization of persuasive moves across negotiation phases. Her broader academic interests include discourse analysis, political linguistics, and innovative, student-centred approaches to language education. Through her work, she aims to connect theoretical linguistic research with practical applications in teaching and communication.